Low-Cost Neural Network Spoof Detection for Unmanned Aerial Systems (UAS)

S26-04

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Acronyms

UAS – Unmanned Aerial System
UAV – Unmanned Aerial Vehicle
NAWCAD – Naval Air Warfare Center, Aircraft Division
LiDAR- Light Detection and Ranging
NN – Neural Network
UE5 – Unreal Engine 5
SDK – Software Development Kit
ML – Machine Learning
RPi – Raspberry Pi
HIL– Hardware in the Loop
SIL– Software in the Loop
CNN– Convolutional Neural Network

Project Overview

Develop a low-cost system to detect spoofed LiDAR signals for UAVs

CUSTOMER REQUIREMENTS

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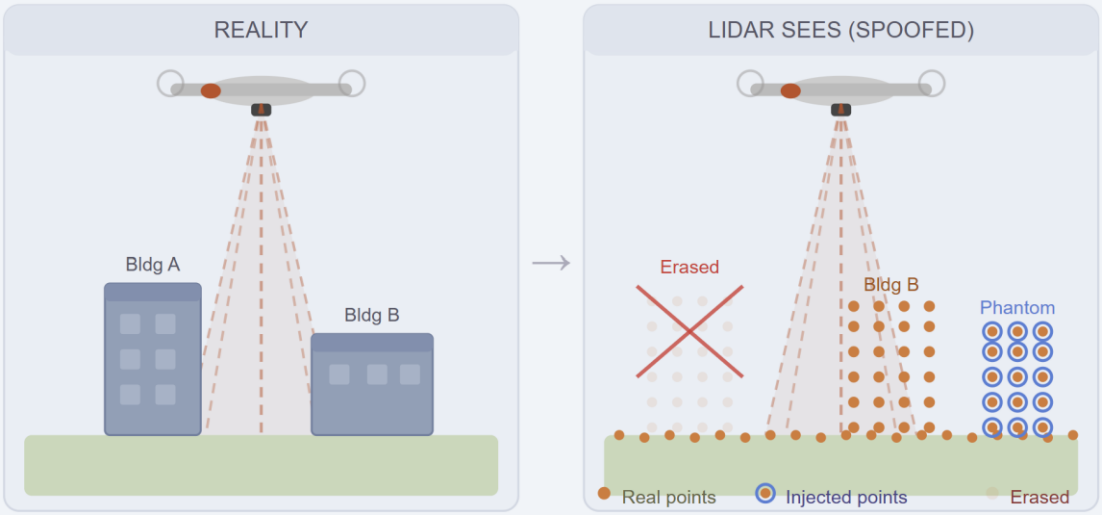
- Build a low-cost spoof detection system for onboard UAV LiDAR sensors
- Use machine learning to distinguish genuine sensor data from spoofed inputs
- Solution must be run on embedded UAV hardware without specialized equipment

OUR APPROACH

- *Simulation* - Unreal Engine 5 virtual drone environment integrated with real hardware (HIL/SIL) for training and validation
- *Detection* - trained ML classification model analyzes live LiDAR data to identify spoofed signals

Background



LiDAR

- Emits laser pulses to build a 3D point cloud of the environment
- Enables UAV navigation, mapping, and obstacle avoidance

Spoofing

- Injects false optical signals into the LiDAR sensor
- UAV perceives fabricated obstacles or terrain, corrupting navigation

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Problem Statement

LiDAR spoofing disrupts how UAVs perceive their surroundings, creating serious risks to navigation and safety. Hardware-based Neural Network detection enables UAVs to reliably identify and reject spoofed signals.

CONSEQUENCES

- **Threatens navigation and safety** — UAV may avoid phantom obstacles or collide with real ones
- **Corrupts terrain mapping** — environmental intelligence becomes unreliable
- **Hard to detect onboard** — spoofed data looks identical to legitimate sensor returns

High Level Overview

High Level Requirements

Our system must:

- Generate a framework to create and simulate spoofed LiDAR signals
- Detect and analyze spoofed signals
- Log and process sensor input data
- Utilize Neural Network with hardware-based inputs

High Level Constraints

- Low-cost system suitable for real-world deployment
- Time limitations for model training
- Limited compute and memory resources

Project Impacts

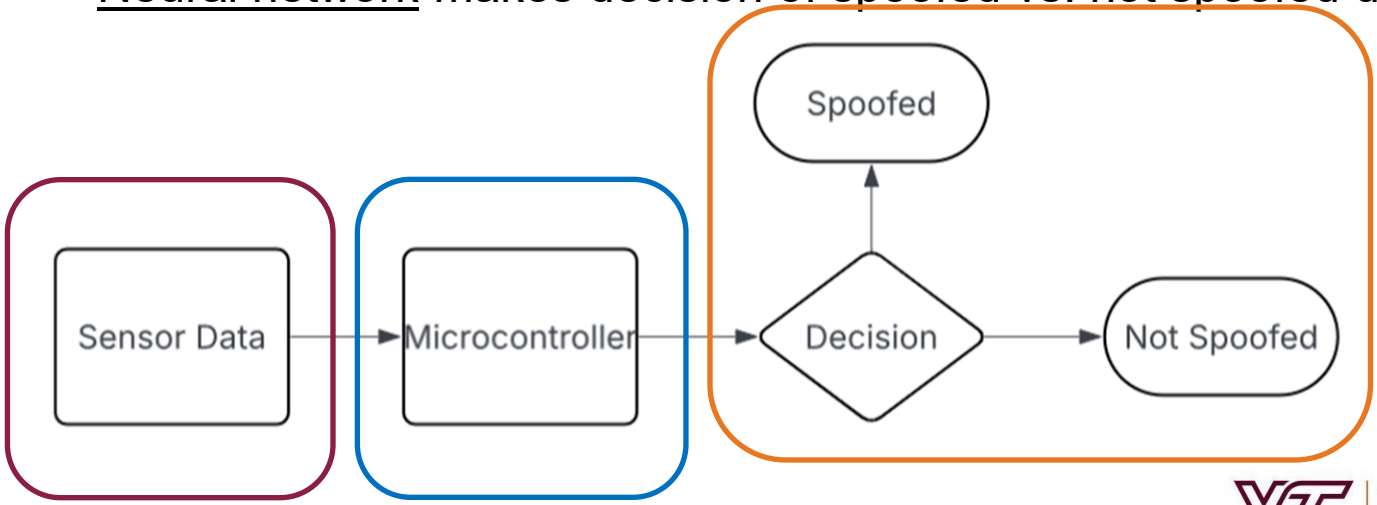
Our Solution Provides:

- The ability to generate LiDAR datasets with a variety of spoofing scenarios for ML model training
- A low-cost, modular spoof detection solution
- Flexibility for different mission systems and goals

View From 20,000 Feet

We created a proof-of-concept system for the following:

- System receives data from the external sensors
- Onboard microcontroller processes data
- Neural network makes decision of spoofed vs. not spoofed data



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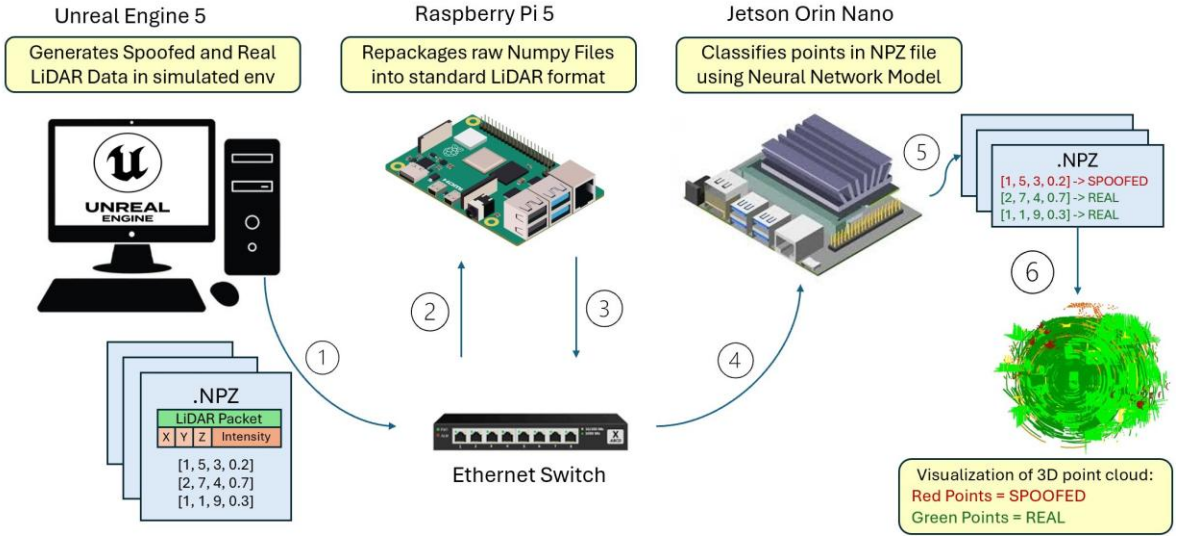
Data Generation – Unreal Engine 5 (UE5)

- Simulation Framework
  - Built on Unreal Engine 5 + Project Air Sim (open source platform)
  - Realistic UAV physics and control
- Sensor & Environment Simulation
  - LiDAR + camera simulation with **noise** and **latency**
  - Fully **customizable** environments (terrain, objects, materials)
- Versatile Data Generation
  - Generates **labeled** perception data for ML pipeline
  - Acts as a **sandbox** for creating adversarial scenarios via material & geometric properties

Data Transfer Architecture



System Steps	
1	UE 5 Data Writing
2,3	RPi Buffer Communication
4	Jetson Packet Transfer
5	ML Classification
6	Neural Network Visual Output

- System **delivers virtual data** from Unreal Engine to the ML model on the Jetson via TCP
- Raspberry Pi provides **data control and manipulation** without sacrificing performance on other subsystems

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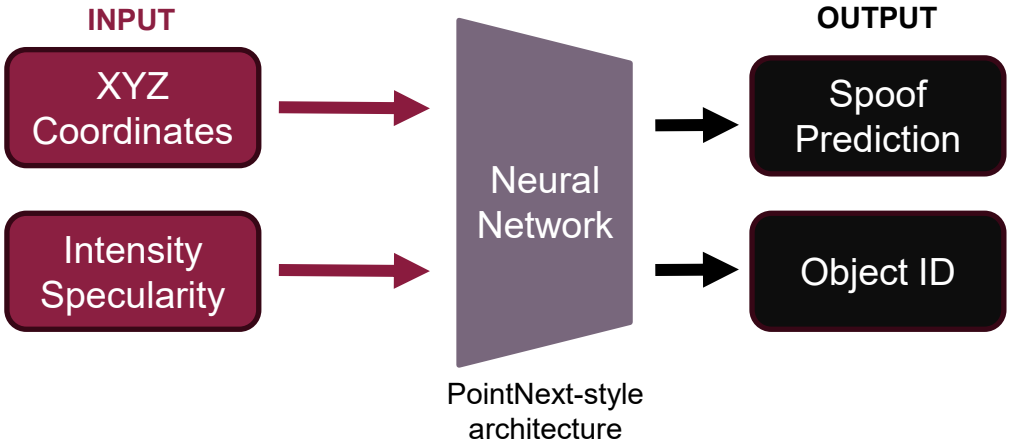
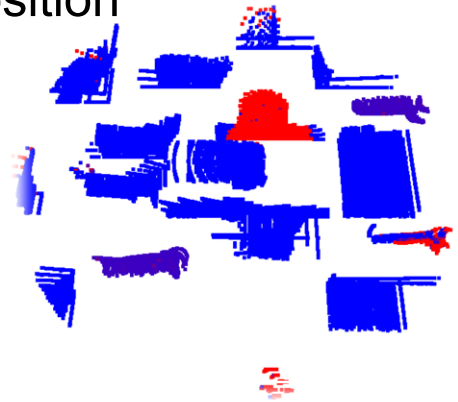
Neural Network Architecture

- Model **segments** point cloud into discrete objects
- **Classifies** objects as spoofed/unspoofed based on material properties (intensity) and position (XYZ coordinates)





Key: Perceived Environment

Real Object

Fake Objects

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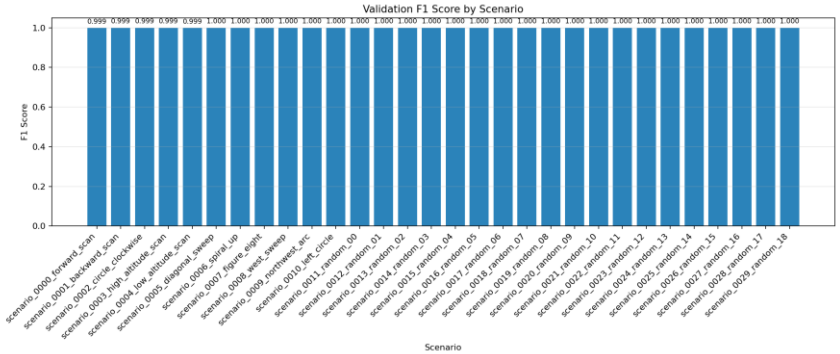
Neural Network Results

- Model trained on **68,977** LiDAR scans across **30** distinct scenarios using supervised learning
- Correctly classified spoofed objects with **99% accuracy**
- Real-time inference rate of **100Hz** from input to output deployed on Jetson Orin Nano



Spoof classification performance by scenario, where spoof classification threshold is above 50%

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Technical Challenges

- Faulty UE5 software pipeline for GPU LiDAR
  - Project AirSim repository error
- Deployment of ML model to Jetson
  - Required re-compilation and extensive rewrite of model code
- Physics based normalization
  - Raw intensity variables with distance

In Summary

We simplified a **complex**, **time consuming**, and **expensive** problem through modeling, creating a solution for **under $300**

- Built **virtual sensing environment** using UE5, LiDAR generation workflows, and communication paths
- Utilized virtual data to **"trick"** our Neural Network into thinking it's on a real drone
- **Generated real and spoofed LiDAR data** to train the Neural Network
- **Tested Neural Network in real-time** resulting in high accuracy

Q & A

Thank you!

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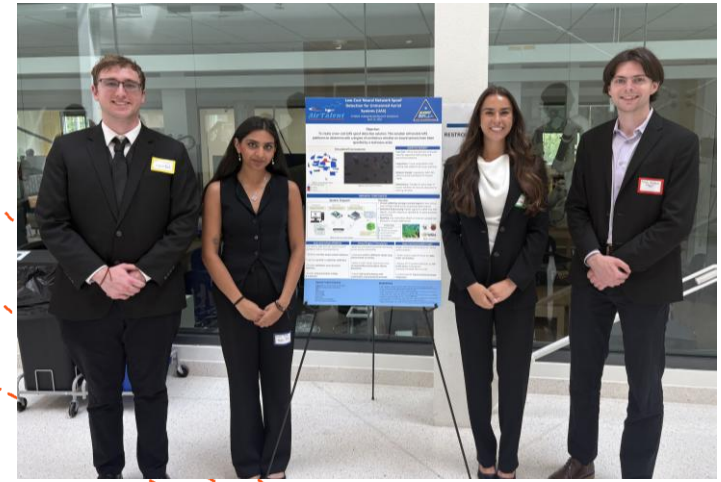

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